
Can Causal Forests reveal hidden dimensions of heterogeneity?

An Application to Gun Ownership and Violence in the United States

1. Introduction

Understanding how different individuals or groups may respond to an intervention is crucial for determining the scale and targeting of a policy’s rollout. In Linear and Partial Linear Models, candidate dimensions of heterogeneity must be specified a priori. Each interaction must be investigated manually, meaning exploration is often limited to standard drivers, such as race and gender. Machine Learning methods such as Causal Forests (CFs) require no pre-specification of variables, meaning they have the potential to identify previously unexplored dimensions of heterogeneity, enabling policymakers to detect and limit unintended impacts. The conditions under which this opportunity is realisable and outweighs the loss of interpretability and increased complexity are undetermined.

To examine this, I employ research on the relationship between gun ownership and violence. In the United States, a person dies from a firearm every 12 minutes on average ([Center for Gun Violence Solutions, a](#)). The rate of gun-related deaths increased by more than 30% between 2014 and 2023, from approximately 10.3 to 13.7 deaths per 100,000 people ([Center for Gun Violence Solutions, b](#)). Gun violence impacts not only its direct victims and perpetrators, but can also diminish children’s educational attainment, worsen maternal health and discourage investment in communities from both households and businesses ([Cook, 2020](#)).

Given U.S. civilians account for an estimated 46% of the worldwide total of civilian held firearms ([del Frate et al., 2019](#)), much attention has been directed towards determining whether the heightened levels gun ownership dissuade violence by acting as deterrent, or perpetuate it. Internationally, a “significant positive correlation between guns per capita per county and the rate of firearm-related deaths” is clear ([Bangalore & Messerli, 2013](#)). However, this may be confounded by country-level characteristics and the diversity in firearm policy more broadly.

In a landmark paper, [Cook & Ludwig \(2006\)](#) estimate the marginal social cost of gun ownership in the United States, quantifying the elasticity of homicide rates with respect to gun ownership as being “between +0.1 and +0.3”. Their major contribution involved applying an improved proxy measure for gun ownership, which has a lower risk of measurement error than survey-based data. [Chernozhukov et al.](#)

(2022) apply a variety of Machine Learning methods to [Cook & Ludwig’s](#) data, finding similar estimates, although model performance varies.

Of key interest to policymakers may be the potential heterogeneity in the effect of gun ownership on violence, given the wide disparity in socio-economic characteristics and access to firearms across the country. Given how controversial restrictions to gun ownership in the U.S. are, any causal claims should be robust and justifiable, before they enter the policy debate.

In the remainder of this paper, I describe the data and methods I use to explore the potential for CFs to reveal heterogeneity, before presenting my results. I offer a discussion on the costs and benefits of applying different methods to identify heterogeneous treatment effects and conclude with a forward look. All tables are listed in [Section 7](#).

2. Data

I source data from [Chernozhukov et al. \(2022\)](#), who adapt the approach of [Cook & Ludwig \(2006\)](#) to assess the capabilities of various Machine Learning techniques for identifying the average treatment effect of gun ownership on homicide rates. [Chernozhukov et al. \(2022\)](#) provide panel data for 195 U.S. counties between 1980 and 1999. I construct two sub-samples using the largest 100 and largest 50 counties¹.

Gun ownership is unobserved. Estimates obtained from survey data are not always available at a sufficiently granular geographic level. They are also unlikely to capture the full scope, as respondents have an incentive to withhold information or simply not participate if they own guns illegally. [Cook & Ludwig \(2006\)](#) employ “the fraction of suicides committed with a firearm (FSS)” as a proxy. They acknowledge FSS still measures gun ownership with error, but argue that it is sufficiently consistent for estimating gun prevalence over time at the U.S. county-level (as long as populations are high enough).

To replicate the approach of [Cook & Ludwig \(2006\)](#), [Cher-](#)

¹[Chernozhukov et al. \(2022\)](#)’s dataset does not include raw population data. They reconstruct population weights, which I use to calculate the average population for each county over the entire period. I select the largest 100 and 50 counties to replicate [Cook & Ludwig \(2006\)](#), who address measurement error bias by limiting their sample to larger counties.

nozhlukov et al. (2022) provide the outcome variable of the log homicide rate by county over time and a large number of controls, including:

- **Crime Controls** - including the robbery rate and burglary rate;
- **Demographic Controls** - including measures of population density (newdens), the Black population share (newblack) and the male population share (newmal);
- **U.S. Census Bureau Controls** - including county-level features of geography, the age distribution, the income distribution, crime rates, housing markets, federal spending, educational attainment, employment and voting patterns.

All variables are weighted by population and logged.

3. Methodology

I construct three models of the effect of gun ownership on homicide rates, with increasing flexibility. In all models:

- $Y_{i,t}$ or $\text{Ln}(Hom)$ = the outcome: log homicide rate in county i at time t
- $D_{i,t-1}$ or $\text{Ln}(FSS)_{t-1}$ = the treatment: log FSS, a proxy measure for gun ownership, lagged by one period
- $X_{i,t}$ = the controls
- $\varepsilon_{i,t}$ = the unobserved error term

The treatment variable is lagged to account for reverse causality. Gun ownership may increase in response to increased levels of violence, whilst also driving the increase in violence itself. Introducing a lag closes the undesired channel of influence of violence driving gun ownership, retaining only the desired channel of gun ownership influencing violence.

3.1. Linear benchmark

The most restrictive specification regresses log homicide rates on log FSS linearly.

$$Y_{i,t} = \beta D_{i,t-1} + X_{i,t} + \varepsilon_{i,t} \quad (1)$$

Under the assumption of conditional independence, β captures the average treatment effect of gun ownership on gun violence, where a 1% increase in the FSS leads to a $\beta\%$ increase in the homicide rate.

To enable heterogeneous treatment effects, I adapt the functional form to:

$$Y_{i,t} = \beta D_{i,t-1} + \delta_k D_{i,t-1} \times W_{k,i,t} + X_{i,t} + \varepsilon_{i,t} \quad (2)$$

where $W_{k,i,t}$ is a pre-specified variable of interest for heterogeneity.

In the fully linear form, I specify three candidates as potential drivers of heterogeneity:

- **newdens** - population density
- **newmal** - male population share
- **newblack** - Black population share

I expect higher rates of urbanisation (higher newdens) amplify the effects of gun ownership on violence when compared to more rural areas (lower newdens), given that higher rates of violent crime are reported in more urbanised areas (Federal Bureau of Investigation). I also expect areas with a higher male population share to have larger effects of gun ownership on violence, given males “account for 86% of all victims of firearm death” (U.S. Centers for Disease Control and Prevention) and U.S. law enforcement agencies “overwhelmingly report a greater percentage of male gang members versus female gang members” (National Gang Center).

I include the Black population share as a third dimension of heterogeneity, but acknowledge that racial composition in the U.S. is inherently tied to other socio-economic characteristics, due to the history of segregation. Inference is conducted with the understanding that the demographic distribution is highly correlated with income and poverty measures.

Each dimension assesses whether the treatment effect shifts by county type, not individual units. For example, when $W_k = \text{newmal}$, δ_k measures the how the overall effect of gun ownership on violence changes for counties of varying male population shares. When $\beta > 0$, $\delta_k > 0$ amplifies the treatment effect, whereas $\delta_k < 0$ dampens it. The reverse is true for $\beta < 0$.

I estimate the specification with each interaction term included individually, and a final specification where all interactions are included simultaneously, denoted by:

$$Y_{i,t} = \beta D_{i,t-1} + \sum_k \delta_k D_{i,t-1} \times W_{k,i,t} + X_{i,t} + \varepsilon_{i,t} \quad (3)$$

where $k = [\text{newdens}, \text{newmal}, \text{newblack}]$.

3.2. Partial Linear Model (PLM)

In the PLM, I relax the linear restriction on the controls, as in Equation (4).

$$\tilde{Y}_{i,t} = \beta \tilde{D}_{i,t-1} + \delta_k \tilde{D}_{i,t-1} \times \tilde{W}_{k,i,t} + g(X_{i,t}) + \varepsilon_{i,t} \quad (4)$$

where $\tilde{A} = A - E(A|X)$ for $A = [Y, D, W_k]$, representing the residualised variables.

$E(Y_{i,t}|X_{i,t})$, $E(D_{i,t}|X_{i,t})$ and $E(W_{k,i,t}|X_{i,t})$ are estimated using a 5-fold implementation of boosted trees with a maximum depth of 4, identified to be the best machine learning model in this setting by Chernozhukov et al. (2022). They are used to residualise Y , D and W_k before conducting the second stage linear regression.

The PLM still requires pre-specification of candidates for heterogeneity. As the outcome, treatment and interaction variables are residualised prior to estimating the treatment effects, only variables which maintain a sufficient level of variation after residualisation are conducive to identifying heterogeneous treatment effects. Of the three candidates, only the male population share has sufficient variation and low enough collinearity to enable heterogeneity analysis.

3.3. Causal Forests (CFs)

Finally, I implement CFs to estimate a fully flexible functional form. CFs extend the framework of Random Forests to focus on maximising the difference between outcome and treatment variable splits, in order to identify a treatment effect. For effects to be causal, the traditional justifications of conditional independence are still required. However, the method enables exploration of heterogeneity that is not pre-determined.

I set the number of estimators to 2000, the minimum leaf size to 5 and extract the ten most important features for discussion.

4. Results

4.1. Linear benchmark

Linear regression results are displayed in Table 1, Table 2 and Table 3. On average, increased gun ownership (measured by proxy through FSS) appears to increase homicide rates. The coefficients sit within the range of elasticities identified by Cook & Ludwig (2006). The positive relationship corroborates research from Bangalore & Messerli (2013).

Contrary to expectations, all interaction terms display a dampening effect. This implies that an increase in population density, male population share or Black population share lessens the impact of increased gun ownership on violence. In the Equation (2) specification, no coefficients are significant, even at the 10% level. When all candidates are included simultaneously as in Equation (3), the direction of effects for newdens and newmal switch, especially in smaller samples. This may be due to the collinearity between variables.

4.2. PLM

Table 4 contains results from the PLM, considering only the male population share as a candidate for heterogeneity. Again, a dampening effect is detected, though estimates lack statistical significance. As sample size decreases, the shrinkage in coefficients switches the average treatment effect of FSS on homicide rates from statistically significant and positive when all 195 counties are used, to statistically insignificant and negative when only the largest 50 counties are used, indicating sample size biases.

4.3. CFs

Table 5 presents the distribution of the average treatment effect as estimated by CFs, across the three samples. The estimates are lower than in the parametric models, although a naive statistical test assuming independence implies that the differences are not statistically significant. Estimates remain positive, even as the sample size decreases, although they lose significance. Given that the PLM estimates tend to sit between those from the Linear Benchmark and the CFs Model, the difference in values may be attributable to the relaxation of restrictions on the functional form, rather than an indication that the Linear model overestimates the treatment effect.

Table 6, Table 7 and Table 8 list the ten most important features identified by the model applied to all 195, the largest 100 and the largest 50 counties respectively. The Black population share is the only pre-specified dimension of heterogeneity that appears in the top ten most important features for any sample size. Best linear predictor tests (presented in Table 9) suggest that the three pre-specified candidates account for 27% of the heterogeneity identified by the Causal Forests model when applied to all 195 counties (indicated by the R-squared). This increases to 44% and 48% for the largest 100 and largest 50 counties respectively. The Black population share continues to be a consistent dampener of the treatment effect, indicating that counties with a higher Black population share experiencing increased gun ownership see a lesser effect on homicide rates compared to counties with a lower Black population share. Population density and male population share estimates do not have a stable relationship when included alongside the Black population share, due to multicollinearity, but univariate estimates also suggest a dampening effect.

The most important feature by far identified by the CFs Model is the outcome variable itself. County-level homicide rates have the greatest importance for heterogeneity, double that of the next most important feature across all three samples. Counties with higher homicide rates experience systematically lower treatment effects compared to counties with lower rates, as indicated by the negative coefficients in the Best Linear Predictor tests in Table 10. The CFs Model

also identifies the distribution of building permits, geographical characteristics and voting patterns as important factors in determining the treatment effect. None of these dimensions were considered in my pre-specification or discussed in the original paper by Cook & Ludwig (2006).

5. Discussion

The results show that CFs can reveal dimensions of heterogeneity that are not pre-specified. It challenges traditional dimensions considered in parametric research on gun ownership and violence by identifying the most important factor as homicide rates. This knowledge is beneficial for policy-makers, as it implies counties with currently low homicide rates may experience a greater social cost from increasing gun ownership than counties where homicide rates are already high. Targeting policies based on pre-existing homicide rates may be more impactful than based on population density or male population share.

The importance of homicide rates is not sensitive to sample size. However, the direction of the impacts of the pre-specified candidates are. The signs for population density and male population share switch as sample size decreases. The CFs Model appears less able to determine the direction of effects when multicollinearity is high, although this holds for the PLM and Linear Model as well.

As the sample size diminishes, observations are restricted to only the largest counties, meaning the data become more homogeneous. The pre-specified candidates then explain a larger portion of the data's heterogeneity. This indicates the benefits of non-parametric approaches reduce as data become more homogeneous or as sample size decreases.

The application of non-parametric approaches also reduces interpretability. The CFs model presents lower average treatment effects and weak heterogeneity effects from the pre-specified candidate variables, but the extent to which these results are driven by the data or the model specifications is unclear. Given the polarising nature of gun ownership legislation in the U.S., the inability to explain how results are achieved and what drives effects discourages the adoption of non-parametric results for determining policy.

PLMs are often considered a middle-ground between the simplicity and interpretability of linear methods and the complexity and "black-box nature" of fully non-parametric approaches. However, for identifying heterogeneity, the PLM appears to reflect the "worst of both worlds". Although it gains flexibility in functional form for the controls, it loses interpretability, whilst still requiring pre-specification of candidate variables, whose set is limited by the residualisation process. My results imply that attempting to balance the interpretability of linear methods with the detective power of non-parametric approaches by implementing a PLM is

not conducive to identifying heterogeneity, especially when trying to reveal hidden dimensions.

It is important to state that all approaches require conditional independence. Employing PLMs or CFs does not gain identification, nor lose it, so long as the implementation separates model generation from treatment estimation. In this setting, the potential measurement error given the use of a proxy treatment variable is retained in all models. The issue is not worsened or resolved by employing Machine Learning methods.

6. Conclusion

The CFs Model is able to reveal hidden dimensions of heterogeneity. Fully non-parametric approaches appear to add the most value when data are heterogeneous. When data are more homogeneous or only small samples are available, the Linear model with well-grounded pre-specified candidates for heterogeneity may be sufficiently effective. PLMs appear the weakest, losing interpretability without gaining new insight into potential drivers of heterogeneous treatment effects.

In the case of gun ownership and violence, the CFs Model reveals that pre-existing homicide rates are of greater importance than population density, male population share and Black population share. Counties with lower pre-existing homicide rates may be susceptible to larger increases in violence as gun ownership increases. This should be considered when determining U.S. gun ownership policy. Future research may focus on understanding the drivers of the dampening effects, especially for population density and male population share, whose effects counter expectations based on the statistical correlation between urbanisation, male population share and violence.

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- Disclaimer:** AI was used in this project to support code development.

7. Annex

Results tables are presented below. Parentheses contain heteroscedasticity-robust standard errors. *, **, *** denotes significance at the 10%, 5% and 1% levels respectively.

7.1. Linear Regression Results

	Ln(Hom)	Ln(Hom)	Ln(Hom)	Ln(Hom)	Ln(Hom)
Ln FSS (t-1)	0.1793*** (0.0652)	0.3306*** (0.1209)	0.2847 (0.1986)	0.2865*** (0.0940)	0.1257 (0.1843)
newdens		-0.0427 (0.0268)			0.0933 (0.0802)
newmal			-0.0981 (0.3172)		-0.1107 (0.0983)
newblack				-0.0839 (0.053)	-0.1180 (0.0836)

Table 1. **Linear Regression Results** using all 195 counties. The first column refers to Equation (1) with no interactions. The next three specifications include each candidate interaction individually, as in Equation (2). The final column refers to Equation (3) where all candidates are included simultaneously.

	Ln(Hom)	Ln(Hom)	Ln(Hom)	Ln(Hom)
Ln FSS (t-1)	0.1237 (0.1299)	0.1023 (0.1004)	0.1309 (0.1087)	0.0532 (0.1902)
newdens	-0.0145 (0.0198)			0.0298 (0.0855)
newmal		-0.0286 (0.0335)		0.0423 (0.1616)
newblack			-0.0393 (0.0340)	-0.1001 (0.0882)

Table 2. **Linear Regression Results** using the 100 largest counties. The first three specifications include each candidate interaction individually as in Equation (2), and the last includes all candidates simultaneously as in Equation (3).

	Ln(Hom)	Ln(Hom)	Ln(Hom)	Ln(Hom)
Ln FSS (t-1)	0.0851 (0.1964)	0.0724 (0.1446)	0.1129 (0.1602)	-0.0214 (0.2765)
newdens	-0.0101 (0.0252)			0.0492 (0.0942)
newmal		-0.0226 (0.0384)		0.0364 (0.1727)
newblack			-0.0353 (0.0372)	-0.1154 (0.1015)

Table 3. **Linear Regression Results** using the 50 largest counties. The first three specifications include each candidate interaction individually as in Equation (2), and the last includes all candidates simultaneously as in Equation (3).

7.2. Partial Linear Model Results

N	195 Ln(Hom)	100 Ln(Hom)	50 Ln(Hom)
Ln FSS (t-1)	0.2113*** (0.0667)	0.0700 (0.0628)	-0.0184 (0.0941)
newmal	-3.8717 (2.5630)	-2.2472 (2.3577)	-2.6834 (3.9639)

Table 4. **PLM Results**, considering only the male population share (newmal) as a candidate for heterogeneity. The columns display regression results using all 195, the largest 100 and largest 50 counties respectively.

7.3. Causal Forests Results

N	tau_mean	tau_std	tau_upper	tau_lower
195	0.0901	0.0957	0.2307	-0.0496
100	0.0926	0.1010	0.2000	-0.0147
50	0.1561	0.1662	0.2668	0.0448

Table 5. **CFs Results** across all three sample sizes (denoted by the first column). The table contains the mean, standard deviation and upper and lower bounds for the average treatment effect.

Variable	Description	Importance
logghomrJ	Homicide Rate	0.1839
logghomr0	Homicide Rate	0.0896
newblack0	Black population share	0.0447
BPS030DJ	Building permits	0.0316
LND110D	Geography	0.0221
BPS030D0	Building permits	0.0213
newblackJ	Black population share	0.0172
newblack	Black population share	0.0172
BPS020D	Building permits	0.0166
ELE065D0	Voting patterns	0.0143

Table 6. **CFs 10 Most Important Features** using all 195 counties.

Variable	Description	Importance
logghomrJ	Homicide rate	0.3028
logghomr0	Homicide rate	0.1341
newblack0	Black population share	0.0512
newblack	Black population share	0.0295
newblackJ	Black population share	0.0195
logrobr	Robbery Rate	0.0145
logrobrJ	Robbery Rate	0.0140
ELE035D	Voting Patterns	0.0084
logburg	Burglary rate	0.0079
BPS120D	Building permits	0.0078

Table 7. **CFs 10 Most Important Features** using the largest 100 counties.

Variable	Description	Importance
logghomrJ	Homicide rate	0.2614
logghomr0	Homicide rate	0.1322
newblack	Black population share	0.0465
newblack0	Black population share	0.0454
newblackJ	Black population share	0.0410
logrobr	Robbery Rate	0.0209
BNK010D	Finance	0.0103
HSG350D	Housing distribution	0.0100
BZA210D	Building permits	0.0096
PIN020D	Income	0.0087

Table 8. **CFs 10 Most Important Features** using the largest 50 counties.

N	195	100	50
Intercept	0.1387*** (0.0004)	0.0921*** (0.006)	0.1827*** (0.01)
newblack	-0.0429*** (0.002)	-0.0666*** (0.002)	-0.0909*** (0.005)
newdens	-0.0034* (0.002)	0.0367*** (0.003)	0.0495*** (0.004)
newmal	0.007* (0.004)	-0.0299*** (0.005)	-0.0534*** (0.01)

Table 9. Best Linear Predictor Results using only pre-specified candidate variables for heterogeneity. The R-squared is 0.265, 0.440 and 0.484 for each of the three samples, in decreasing sample size.

N	195	100	50
Intercept	0.1717*** (0.005)	0.1704*** (0.005)	0.3835*** (0.006)
newblack	-0.0221*** (0.002)	-0.0252*** (0.002)	-0.0008 (0.002)
newdens	-0.0136*** (0.002)	0.0173*** (0.002)	0.0115*** (0.001)
newmal	0.0176*** (0.004)	-0.0181*** (0.003)	-0.0199*** (0.003)
logghomrJ	-0.0276*** (0.002)	-0.0566*** (0.002)	-0.1620*** (0.003)

Table 10. Best Linear Predictor Results using the pre-specified candidate variables for heterogeneity and log homicide rates, as identified by the CFs Model. The R-squared is 0.306, 0.589 and 0.879 for each of the three samples, in decreasing sample size.